

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: I am looking for a sectional terminal block

Assistant: To better assist you, could you please clarify:

1. What is the exact use of the polymer in your sectional terminal block? For example, is it for the housing, cover, shell, or another component?
2. Where will this terminal block be used in your solution? Is it for an industrial control panel, railway application, photovoltaic system, or another end-use?

For instance, terminal block housings can be used in industrial energy systems, railway control panels, or photovoltaic installations. Understanding your specific scenario will help identify the most suitable material options.

User: housing and insulation. It will see service temperature up to 130C

Assistant: Thank you for clarifying the use for housing and insulation, and the service temperature requirement.

Could you please provide the following details:

1. What is the operating environment for the terminal block? For example, will it be installed indoors in a protected enclosure, or exposed to vibration, dust, humidity, or outdoor conditions?
2. What is the minimum wall thickness of your housing/insulation part?

3. Do you have any preferred base materials for this application (such as PA, PBT, PC, etc.), or are you open to alternative material suggestions?
4. Are you open to alternative material suggestions if your preferred material is not available or optimal for your requirements?

These details will help ensure the material suggestions are tailored to your needs.

User: 0.5 mm thickness and dont have preferred material

Assistant: Here are the high-confidence material requirements (CTQs) assumed for your sectional terminal block housing and insulation:

1. Minimum wall thickness: 0.5 mm
2. Flame rating: UL 94 V-0 at 0.5 mm
3. Glow Wire Flammability Index (GWFI): 960°C at 0.5 mm
4. Glow Wire Ignition Temperature (GWIT): 750–775°C at 0.5 mm
5. Comparative Tracking Index (CTI): 600 V
6. Relative Thermal Index - Electrical (RTI Elec): 130°C
7. Injection molding required
8. Flame retardant feature required
9. Non-halogenated/red phosphorus-free flame retardant preferred
10. Dimensional stability required
11. Low moisture absorption preferred

Are all of these requirements correct for your application? Please confirm so I can proceed to refine the remaining material requirements.

User: It needs to meet UL1059

Assistant: Thank you for confirming the requirements and specifying UL1059

compliance.

To further refine the material selection, could you please clarify the following:

1. Do you require glass fiber reinforcement in the polymer (for added mechanical strength and dimensional stability), or is an unfilled material sufficient?
2. Is there any need for resistance to soldering heat or chemicals (for example, if the terminal block will be near soldering operations or exposed to cleaning agents)?
3. Is there any specific certification footprint required beyond UL1059 (such as VDE, railway EN 45545, or others)?

Your answers will help ensure the material suggestions are fully aligned with your application needs.

User: i dont have anything you can recommend the grades

Assistant: Based on your requirements for sectional terminal block housing and insulation (UL1059 compliance, 0.5 mm wall, flame retardancy, high CTI, glow wire indices, and thermal stability), the following grades may be suitable options:

1. CELANEX® XFR 6852 GF30 - Excellent dimensional stability, low moisture absorption, UL 94 V-0 at 0.4 mm, CTI 600 V, GWIT $\geq 775^{\circ}\text{C}$, RTI Elec 140°C . Purpose-built for thin-wall electrical connectors and terminal blocks.
2. FRIANYL® XT4 GF30 V0I NC 1102/F - High thermal stability (PPA), UL 94 V-0 at 0.4 mm, CTI 600 V, GWFI 960°C , GWIT 800°C , RTI Elec 140°C . Suitable for thin-wall, high-temperature applications.
3. Zytel® FR70G30V0NH NC010 - Robust PA66 platform, UL 94 V-0 at 0.4 mm, CTI 600 V, GWFI 960°C , GWIT $750\text{--}800^{\circ}\text{C}$, RTI Elec 140°C . Widely used for electrical housings

and connectors.

4. Zytel® HTNFR42G30NH NC010 – PPA with strong thermal capability, UL 94 V-0 at 0.4 mm, CTI 600 V, GWFI 960°C, GWIT 750–825°C, RTI Elec 130°C.

5. FRIANYL® A3 GF30 V0 – PA66, UL 94 V-0 at 0.4 mm, CTI 600 V, GWFI 960°C, GWIT 775–800°C, RTI Elec 130°C. Strong historical usage in terminal blocks.

6. FRIANYL® A3 GF25 V0 NC 1102 – PA66, 25% GF, UL 94 V-0 at 0.4 mm, CTI 600 V, GWFI 960°C, GWIT 775–800°C, RTI Elec 130°C.

These grades are potential options based on technical data and historical success in terminal block applications. Please ensure final material selection and product design decisions are validated for your specific requirements and compliance with all applicable standards and regulations. Celanese does not manufacture or certify products for UL1059 applications; always confirm suitability with your testing and certification process.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.